Image-based Trajectory Determination of Artificial Satellites

Preliminary outline on geometry, data acquisition and determination models

Frederic Heil
frederic.heil@hs-weingarten.de
Ravensburg-Weingarten University of Applied Sciences

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1 Introduction

Image-based trajectory determination of artificial satellites using ground based cameras is a multi-stage procedure that, in order to be implemented successfully, requires several problems to be solved. A thorough understanding of the underlying geometry and reference frames is needed. Once that has been established, methods for obtaining visual data can be defined, specifically for position and view vectors. These serve as an essential input for trajectory determination algorithms which will be used to determine the properties of a Keplerian orbit and its dedicated orbital plane.
The scope of this paper is limited to ground based observations, specifically systems using one or two cameras which are not supposed to change their location on Earth for the process of imaging. No additional technologies, such as radar or laser measurements are to be used. Camera calibration in this context, specifically the process of determining view vectors, initially uses a star-based calibration procedure, which limits the field-of-view range of cameras to certain values, to be elaborated in section 3.3. Furthermore, the automated distinction between stars and satellites in measurement images could be considered as an additional problem to be solved, but is not part of the scope.
Trajectory-model-wise, the scope of this paper is limited to Keplerian orbits, which solve the 2-body problem with point masses. For the observable objects, namely artificial satellites, the main focus lies in the region of Low Earth Orbit. Planet Earth is the main gravitational dominating force, observation arcs are rather short and because of this, more complicated orbital models in terms of n-body-physics are simply infeasible, thus not part of the scope.

2 Geometry

2.1 The Keplerian orbit

Geometrically, the shape of a Keplerian orbit can be described as an intersection between plane and cone, which carries some implications. Firstly, the shape of an orbit is either a circle, an ellipse, a parabola or a hyperbola. Secondly, as the intersection implies, the orbit lies within a plane. [1]
The current orbital radius r, which represents the current distance of an orbiting object (a satellite) from a gravitational center (planet Earth) can be described with the following formula for circles and ellipses in a polar coordinate system, as provided by Weber [3] and modified to account for argument of periapsis ω:

$$r(\theta) = \frac{a(1-e^2)}{1 + e \cdot \cos(\theta - \omega)} \tag{1}$$

Hereby, a represents the semi major axis of the orbit, which corresponds to the average distance between satellite and gravitational center. For elliptical orbits, this is just the

average between the closest distance (periapsis r_p) and the furthest distance (apoapsis r_a), as illustrated in figure 1, stated below in equation 2:

$$a = \frac{r_p + r_a}{2} \tag{2}$$

The parameter e represents the eccentricity, an inherent property of conic sections. For circles it is $e = 0$, so that equation 1 collapses to $r(\theta) = a$. For ellipses it is $0 < e < 1$, for parabolas $e = 1$ and for hyperbolas $1 < e$. The parameter θ represents the so called true anomaly, which simply corresponds to the parameter angle in a polar coordinate system. [2]
The angle ω corresponds to the argument of periapsis, which marks the closest distance (periapsis) in the polar coordinate system.
For hyperbolas, equation 1 is to be slightly modified, such that the sign of the dividend changes, which leads to:

$$r(\theta) = \frac{-a(1-e^2)}{1 + e \cdot \cos(\theta - \omega)} = \frac{a(e^2-1)}{1 + e \cdot \cos(\theta - \omega)} \tag{3}$$

[4]
These type of trajectories posses an asymptote which is never physically reachable by the moving satellite, rather being a mathematical property of the function. The asymptote, which the satellite moves along, is the one with the gravitational center in its focal point. The semi major axis a can also be defined by equation 2, however the apoapsis is located on the unreachable asymptote, as illustrated in figure 2.
As apparent from equation 1, it does not hold up for an eccentricity value of $e = 1$ for parabolas. In case of $e = 1$ a modified equation is provided as follows:

$$r(\theta) = \frac{p}{1 + \cos(\theta - \omega)} \tag{4}$$

Hereby p corresponds to a constant value, so that $\frac{p}{2}$ equates to the distance between gravitational center and periapsis. A semi major axis value a can not be defined as there is no apoapsis for this type of trajectory. [5]
Equations 1, 3 and 4 yield real numbers when plugging in concrete values for θ. They can be expanded into parametric curves, which helps visualize the actual appearance of the different types of trajectories as follows:

$$\begin{aligned} r(\theta) &= \frac{a(1-e^2)}{1 + e \cdot \cos(\theta - \omega)} \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} \\ r(\theta) &= \frac{a(e^2-1)}{1 + e \cdot \cos(\theta - \omega)} \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} \\ r(\theta) &= \frac{p}{1 + \cos(\theta - \omega)} \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} \end{aligned} \tag{5}$$

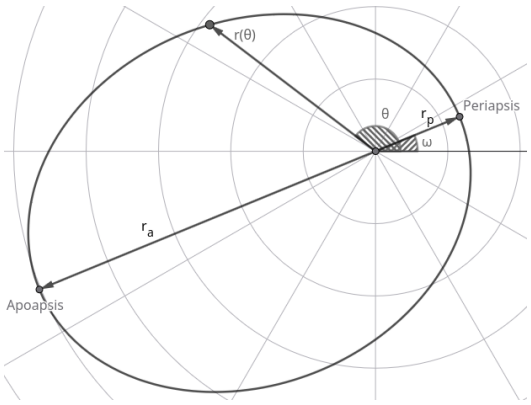


Figure 1: Parametric curve of an elliptic obit.

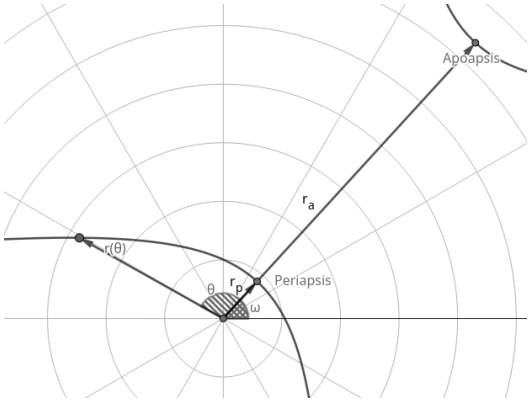


Figure 2: Parametric curve of a hyperbolic trajectory.

This covers the different types of Keplerian trajectories as well as their properties in two dimensions.
Stated above, geometrically, each trajectory can be defined as intersection between cone and plane in three dimensions. This leaves the property of the orbital plane to be defined. This plane always contains the point of gravitational center. Also, the gravitational center shall be defined as the origin of the coordinate system, as elaborated below in section 2.2. Hence, the plane is sufficiently defined by a normal vector n.

2.2 Reference coordinate system

The reference coordinate system of our choice is the Earth Centered Inertial (ECI) coordinate system. This coordinate system is aligned to planet Earth but not fixed to its surface. The Z-axis is aligned to Earth's rotation axis, while the X-Y-plane is aligned to Earth's equatorial plane, as visualized below. Both X-, and Y-axes can be considered fixed in reference to the background stars when dealing with short time periods. For time periods in the range of months and years this is not the case due to precession, as elaborated in section 3.4. Each position can be described by a vector in $\mathbb{R}^3$.

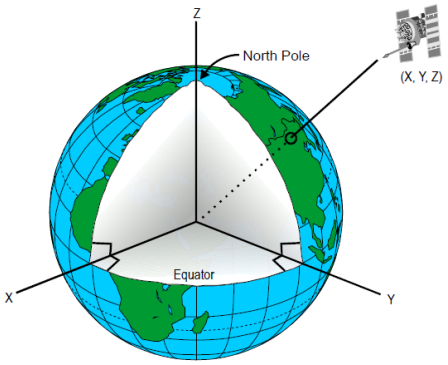


Figure 3: Earth Centered Inertial coordinate system aligned to planet Earth. [6]

In this reference frame, when stationed at a fixed location on planet Earth, the position vector changes over time. This is different from the Earth Centered Earth Fixed (ECEF) coordinate system, which is fixed to Earth's surface, such that position vectors in a fixed position do not change over time. [6]

3 Data acquisition

3.1 Earth-bound camera-based observation

Before setting up mathematical models and implementing trajectory determination algorithms, we need to take a look at what kind of data is readily available, what is unavailable (to be determined by algorithms later on) and how observed images look like. This serves as the basis on how we have to define models and what problems need to be solved. For this purpose, we define the geometric model of a stationary observer / camera on the surface of planet Earth using the following figure:

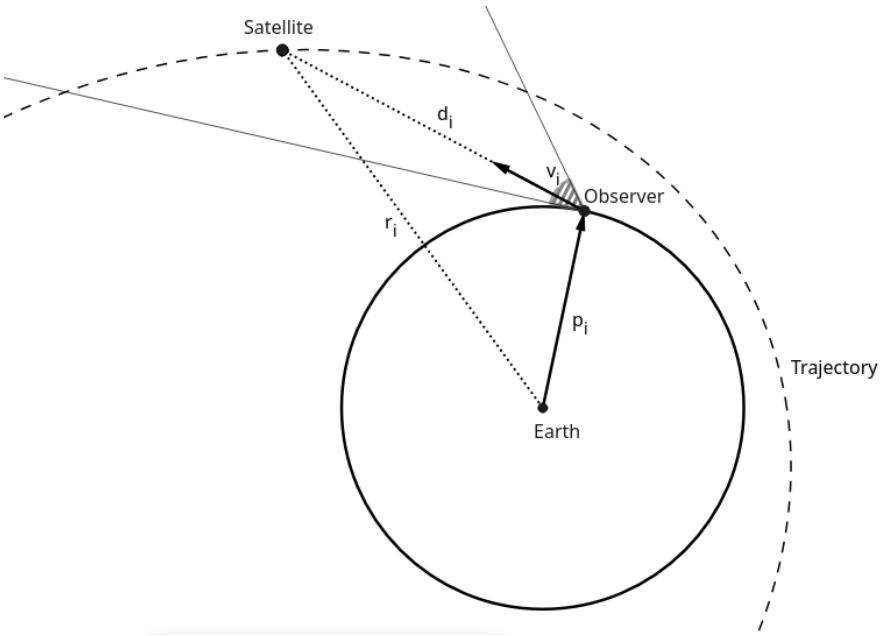


Figure 4: Initially known and unknown properties of Earth-bound camera-based satellite observation depicted in a two dimensional representation.

Hereby, all variables are annotated with an index i. That is, because these values change over time and are only valid for a certain time stamp i, which is a known property.
The first readily available property apart from the time stamp is the position vector $p_i \in \mathbb{R}^3$, which describes the stationary position of the observer on planet Earth in ECI coordinates. The determination of this property is elaborated in section 3.2.
Secondly, we have the view vector $v_i \in \mathbb{R}^3$. This vector marks the visual direction of the satellite as seen from the observer's position in ECI coordinates. The magnitude of this vector does not match the distance between observer and satellite, as this property is yet to be determined. Because of this, the value if the vector is normed to unit length.
Further known or readily determinable properties are camera properties. For one, the camera's field of view is a fixed value. Moreover, the cameras alignment / rotation is readily determinable, so that each pixel can be assigned to a view vector v_i. The process of view vector determination and camera alignment is elaborated in section 3.3.
This sums up all readily available or readily determinable properties of Earth bound camera based satellite observation. The unknown properties of this model comprise of the absolute position vector of the satellite $r_i \in \mathbb{R}^3$ and the distance vector $d_i \in \mathbb{R}^3$ which marks the distance between observer and satellite, of which only the direction is known as v_i. The vectors are related to one another as they form the sum $r_i = p_i + d_i$.
Now, that available and unavailable parameters have been established, we can examine the visual data that is captured as images and what the appearance of satellites will be.
As visible from the ground, satellites appear as more or less bright dots similar to stars, even so that they may be observable to the naked eye. Their movement appears smoothly

and their brightness does not change drastically, as they do not posses flashing lights like aircraft. They are illuminated by the sun, until they enter Earth's shadow at which they become invisible. [7]

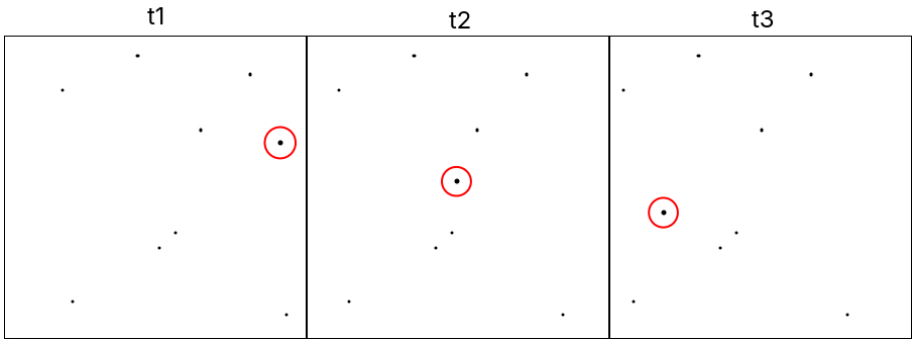


Figure 5: Stars and satellite (encircled) are perceived as dots by cameras.

Illustrated above in figure 5 are three schematic image frames, supposed to be taken at three time stamps t_1, t_2 and t_3 by a ground-fixed camera. In each frame, stars and a satellite (encircled) are visible. For one, the satellite would not be distinguishable from the background stars if it were not marked. Differentiation between satellites and stars may be automated, however we consider this issue not to be part of this research project. Next, there are two relevant types of apparent movements visible in the image series. As planet Earth rotates, background stars move slowly in the image frame. The apparent movement of the satellite is caused mainly by its inherent movement along its trajectory and marginally by the rotation of planet Earth.
These aspects have to be factored in, especially during camera alignment and view vector determination procedures.

3.2 Determination of position vectors

As stated previously in section 3.1, the position vector p_i in ECI coordinates is a readily available value. The actual vector values are initially unknown and have to be derived from known values of the observer. These known values are latitude, longitude and height on Earth and the current time stamp. From these four values, the p_i vector can be sufficiently derived. First of all, we have to factor in the shape of planet Earth, which is ellipsoidal, rather than perfectly spherical. Sanz Subirana et. al. provide a set of equations to perform a transformation from latitude, longitude and height to a $\mathbb{R}^3$ vector in ECEF coordinates, which can be transformed to ECI coordinates later on.
As follows, we define latitude as φ, longitude as λ and height above ellipsoid as h. Further known general values are Earth's equatorial radius a and polar radius b. The geometric context is illustrated in figure 6 below:

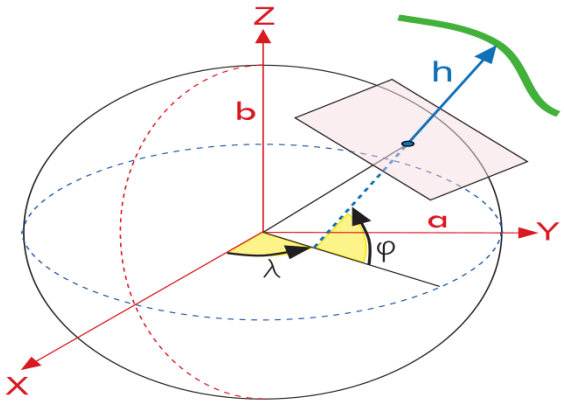


Figure 6: Cartesian and ellipsoidal coordinates on planet Earth. [8]

As the shape of Earth is ellipsoidal on large scale, an ellipse can be defined as a section of the ellipsoid along the Z-axis, for example in the YZ-plane. This ellipse, just as the orbital ellipses, possesses an eccentricity e which is defined as follows:

$$e^2 = \frac{a^2 - b^2}{a^2} \tag{6}$$

Using the value of e, the radius N subject to φ of the prime vertical is defined as follows:

$$N = \frac{a}{\sqrt{1 - e^2 \sin(\varphi)^2}} \tag{7}$$

Using e, N and the height value h, the components of position vector p can now be sufficiently defined as:

$$p = \begin{pmatrix} p_x \\ p_y \\ p_z \end{pmatrix} = \begin{pmatrix} (N+h)\cos(\varphi)\cos(\lambda) \\ (N+h)\cos(\varphi)\sin(\lambda) \\ ((1-e^2)N+h)\sin(\varphi) \end{pmatrix} \tag{8}$$

[8]
In the ECEF coordinate system, the longitude value λ is always fixed. In the ECI coordinate system, Earth rotates around the Z-axis, as such that of the parameters latitude, longitude and height only longitude in reference to the ECI coordinate origin changes as time progresses. To determine longitude subject to time, Curtis provides a set of equations and a process to solve this issue. To elaborate the following process, some underlying geometric elements must be defined beforehand.

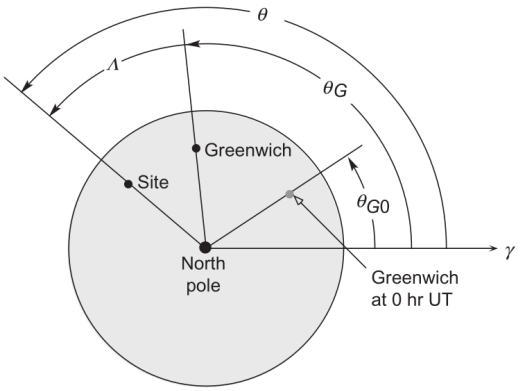


Figure 7: Geometric context of ECI longitude subject to time. [9]

As illustrated above in figure 7, the geometric context consists of several angles along the polar axis, which corresponds to the Z-axis in our ECI coordinate system, as illustrated

in figure 3. The axis γ corresponds to the X-axis. Time stamps in this context are defined as *YYYY.MM.DD HH:MM:SS*, which consists of year, month, day, hours, minutes and seconds to date any observation in Universal Time (UT).
The first relevant value is θ_{G0}, which corresponds to ECI longitude angle at which Greenwich (hereby short for the prime meridian of the ECEF coordinate system) is located on any day at 0hr UT, meaning *YYYY.MM.DD 00:00:00*.
Next, we have θ_G, the angular location of Greenwich at the time stamp of observation, meaning that hours, minutes and seconds in the time stamp are fully specified.
Finally, we have Λ and θ related by $\theta = \Lambda + \theta_G$, where Λ describes the longitude angle between observer and Greenwich at time of observation and θ describes the longitude angle between observer and axis γ. To determine these values, specifically starting with θ_{G0}, a complex process is required which consists of several steps, outlined below:

1. The determination of the current Julian day (J_0):
   The current Julian day depicts the number of days since noon (UT) on January 1st, 4713 BC. For any given date within limitations, consisting of year (y), month (m) and day (d), the following formula applies:

$$J_0 = 367y - \left\lfloor \frac{7\left(y + \left\lfloor \frac{m+9}{12} \right\rfloor\right)}{4} \right\rfloor + \left\lfloor \frac{275m}{9} \right\rfloor + d + 1721013.5 \tag{9}$$

   constrained to $1901 \leq y \leq 2099$, $1 \leq m \leq 12$ and $1 \leq d \leq 31$.

2. The determination of T_0 since the J2000 epoch:
   The J2000 Julian epoch is defined as noon (UT) on January 1st, 2000 and marks the number of days since the start of the Julian calendar, which is 2451545. This step consists of calculating the number of Julian centuries T_0 between J2000 and the current observation date. This can be archived with the following formula containing the result of the previous step:

$$T_0 = \frac{J_0 - 2451545}{36525} \tag{10}$$

3. The determination of θ_{G0}:
   With the result T_0 of the previous step, we can now determine the angle θ_{G0} with the following polynomial, which yields the value in degrees:

$$\theta_{G0} = 100.4606184 + 36000.77004 \cdot T_0 + 0.000387933 \cdot T_0^2 - 2.583(10^{-8}) \cdot T_0^3 \tag{11}$$

4. The determination of θ_G and θ:
   Having calculated θ_{G0}, we can now compute θ_G for any arbitrary time of the day using the following equation, which yields a degree value:

$$\theta_G = \theta_{G0} + 360.98564724^\circ \cdot \frac{UT[\text{hr}]}{24[\text{hr}]} \tag{12}$$

   Hereby UT corresponds to the current time of the day as floating point number (e.g. the time value 09:41:22 corresponds to 9.68944 [hr]). The angular value slightly above 360° stems from the fact, that the rotation of Earth against the background stars, a sidereal rotation, is slightly shorter than 24 hours. With θ_G determined and known Λ value, θ arises from the geometric context as $\theta = \Lambda + \theta_G$.

[9]

Now, that θ, the longitude of the observer in ECI coordinates at a specific time stamp i, has been determined, we can use the previously established equation 8 to determine the position vector. This can be done by inserting θ for λ which yields the final result of p_i.

3.3 Determination of camera alignment and view vectors

This section elaborates the determination of camera alignment and view vectors using a previously established star-based calibration procedure. Hereby a view vector constitutes to the variable v_i as explained in section 3.1. As for the camera alignment, we assume that any observer camera is fixed to the ground (as in fixed to the ECEF coordinate system) and pointing in a certain direction. Furthermore it may be rotated in such a way, that we initially can not deduct any view vectors (in ECI coordinates) from dots in an image.

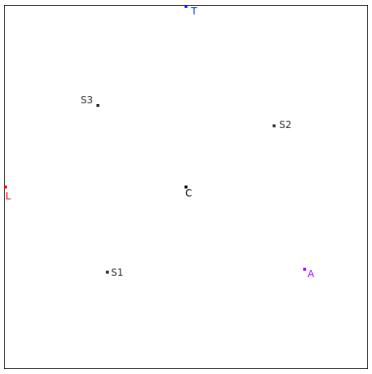


Figure 8: Calibration aspects of an image

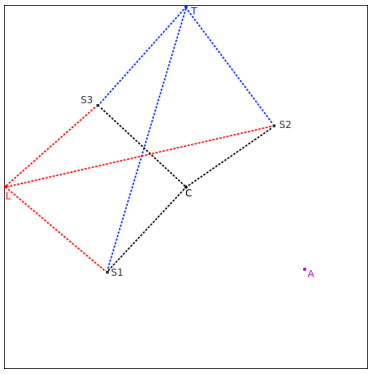


Figure 9: Determination of camera alignment

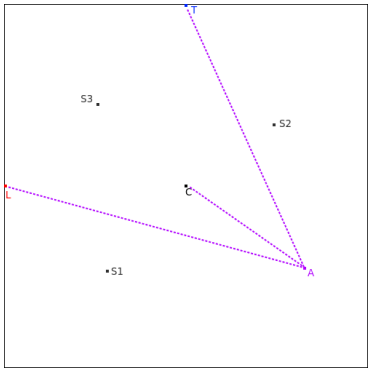


Figure 10: Determination of the view vector

First, we need to determine alignment which constitutes to the calibration of the camera. Illustrated in figure 8 are visual components of an image, taken at a certain time stamp, which are necessary for calibration. Within the image frame, we define three key points: C (located in the center), T (centered on the top edge) and L (centered on the left edge). Visible within the image are dots, which correspond to at least three known stars ($S1$, $S2$ and $S3$) and (initially optional) the artificial satellite A. The assignment of three dots to actual known stars requires a star matching algorithm to be carried out. Such an algorithm has been established in previous work [10] which constitutes to a modified version of the brightness independent 4-star matching algorithm presented by Ying et. al [11].
Once three stars have been identified, three view vectors (in ECI coordinates) v_{S1}, v_{S2} and v_{S3} can be assigned to the three dots $S1$, $S2$ and $S3$.
Figure 9 illustrates the triangulation of the key points C, T and L, a process that yields three corresponding view vectors v_C, v_T and v_L. To triangulate any pixel point P, as in to determine v_P, we have to determine the angular distances between star dots and P first using the following formula:

$$\gamma_i = \delta \cdot \frac{d_i}{w} = \delta \cdot \frac{|S_i - P|}{w} \tag{13}$$

It is assumed that the image is rectilinear and square with a fixed pixel width w and a corresponding angular diameter δ. The value γ_i represents the angular distance between a star point S_i and P, whereas their pixel distance is expressed as d_i. These γ values are to be determined between any P and each of the three stars and are plugged in to the next equation:

$$\begin{bmatrix} — & v_{S1} & — \\ — & v_{S2} & — \\ — & v_{S3} & — \end{bmatrix} \cdot \begin{bmatrix} v_P \end{bmatrix} = \begin{bmatrix} \cos(\gamma_1) \\ \cos(\gamma_2) \\ \cos(\gamma_3) \end{bmatrix} \tag{14}$$

Solving this equation for v_P yields the corresponding vector value (in ECI coordinates) for any point P on the image. [10]
If the artificial satellite A is visible in the same image, v_A can be determined similarly using equations 13 and 14 using the same process. A more generalized approach however is to determine v_A of A by triangulating it with the previously triangulated points C, T and L, as shown in figure 10. Initially after calibration, the satellite might not be visible in the camera's field of view, but at a later point in time. Stated previously, the camera is fixed to the ground, meaning that the observation frame moves as Earth rotates. This means, that the determined image calibration vectors v_C, v_T and v_L which are fixed to the image also move with Earth's rotation. Supposing that v_C, v_T and v_L have been determined at time stamp t_0, we can determine v'_C, v'_T and v'_L at t_1 using the following rotation angle and matrix:

$$\alpha = 360.98564724° \cdot \frac{\Delta t}{T_d} = 360.98564724° \cdot \frac{(t_1 - t_0)[\text{hr}]}{24[\text{hr}]}$$
$$v'_P = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot v_P \tag{15}$$

Stated previously in section 3.2, Earth rotates slightly more than 360° in 24 hours relative to the ECI coordinate system. With the difference $t_1 - t_0$ we obtain the rotation angle α at which the observer location rotates relative to the Z-axis. That angle can be applied to v_P using a rotation matrix which rotates the vector around the Z-axis as well.
As the satellite dot shows up in a later image at t_1, we can now triangulate v_A of A using v'_C, v'_T and v'_L equivalently to the previously performed star calibration of C, T and L.

3.4 Accounting for axial precession of planet Earth

Defined previously in section 2.2, the ECI coordinate system, specifically the Z-axis, is aligned with Earth's rotation axis. Over longer time periods of several months to years this axis can not be considered to be fixed in space as it performs a slow tumbling motion. Over the course of around 26000 years the axis will complete a full precession period so that it is facing in its initial direction again [12] as visualized below:

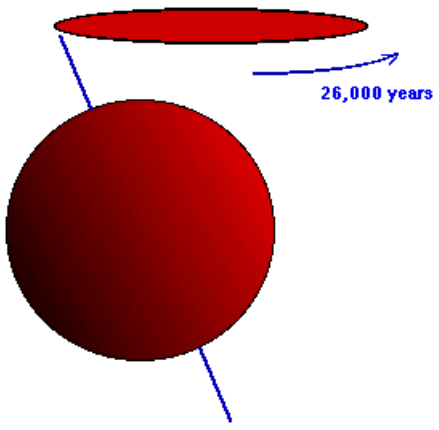


Figure 11: Tumbling movement of Earth's rotation axis [12].

Ignoring other (outside the scope of this paper) long term gravitational effects of the Sun, Moon and Earth's equatorial bulge, the orbits of near Earth satellite's trajectories should remain fixed to the background stars, more specifically the plane normal vector n. These circumstances will cause the vector n and background stars to perform an apparent precession movement in the ECI coordinate system which has to be accounted for when performing star-based camera calibration or comparing measured orbital properties to

ground truth values defined further in the past.
Lieske provides a set of equations to formulate a rotation matrix with which ECI vectors can be multiplied with. It starts with the definition of the initial reference time T and current time t denoted in Julian centuries as follows:

$$\begin{aligned} T &= \frac{JD(\varepsilon_F) - JD(J2000)}{36525} \\ t &= \frac{JD(\varepsilon_D) - JD(\varepsilon_F)}{36525} \end{aligned} \tag{16}$$

Hereby, $JD(\varepsilon_F)$ marks the number of Julian days since J2000 for an initial starting epoch ε_F, whereas ε_D marks the current time epoch. Our implementation of the star calibration procedure uses the BSC5P bright star catalog which defines star view vectors at $\varepsilon_F = J2000$ [14] resulting in a collapse of the initial reference time to $T = 0$.
With the time parameters established, three rotation angles ζ, z and σ are defined by the following polynomials. Resulting angle values are denoted in arcseconds.

$$\begin{aligned} \zeta &= \left(2306.2181 + 1.39656\,T - 0.000139\,T^2\right) t + \left(0.30188 - 0.000344\,T\right) t^2 + 0.017998\,t^3 \\ z &= \left(2306.2181 + 1.39656\,T - 0.000139\,T^2\right) t + \left(1.09468 + 0.000066\,T\right) t^2 + 0.018203\,t^3 \\ \sigma &= \left(2004.3109 - 0.85330\,T - 0.000217\,T^2\right) t - \left(0.42665 + 0.000217\,T\right) t^2 - 0.041833\,t^3 \end{aligned} \tag{17}$$

Next, let $R(\alpha)$ be the rotation matrix to rotate around the Z-axis by angle α and $Q(\alpha)$ the rotation matrix to rotate around the Y-axis by angle α, the precession adjustment for any unprecessed ECI vector v_0 to obtain precessed v is defined as follows:

$$v = R(-z) \cdot Q(\sigma) \cdot R(-\zeta) \cdot v_0 \tag{18}$$

[13]
This precession correction must be applied to plane normal vectors n of ground truth orbits defined further in the past to make them comparable to measurements. Furthermore, all view vectors of stars listed in a star catalog used for calibration must be precessed similarly.

4 Orbit determination models

4.1 Parametric trajectory from angles and radii

Previously, in section 2.1 we have established the Keplerian orbit. Elliptical and hyperbolic orbits can be described using equations 1 and 3, characterized by a, e and ω. These equations differ from each other in the sign, but yield similar curves mirrored at the X-axis if equipped with the same values a, e and ω otherwise. For this reason, equation 1 can also be used to model hyperbolic trajectories, the sign is simply flipped. If we allow negative values specifically for a and e, this issue can be omitted and equation 1 can be used in the following to model both elliptical and hyperbolic trajectories.
We start off with three distinct measurements where three radii r_i and three angles θ_i for $i \in \{1, 2, 3\}$ are known. Geometric context is given in figures 1 and 2. Going back to section 3.1 we know that r_i values are initially unknown. Their determination is an issue for upcoming sections.
We start the derivation process by plugging in radii and angular values into equation 1, which yields a system of three equations for $i \in \{1, 2, 3\}$ structured as follows:

$$r_i = \frac{a(1 - e^2)}{1 + e \cdot \cos(\theta_i - \omega)} \tag{19}$$

From this, we rearrange equation 19 to factor out a subject to e and ω as follows:

$$r_i = \frac{a(1-e^2)}{1+e\cdot\cos(\theta_i-\omega)} \Leftrightarrow a = \frac{r_i(1+e\cdot\cos(\theta_i-\omega))}{1-e^2}$$

$$\boxed{a = \frac{r_i(1+e\cdot\cos(\theta_i-\omega))}{1-e^2}} \tag{20}$$

Next, we obtain e subject to ω by equating expression 20 with itself but with two different index values i and j with $j \in \{1,2,3\}$ and $i \neq j$ to use different radii and angles as follows:

$$\begin{aligned}\frac{r_i(1+e\cdot\cos(\theta_i-\omega))}{1-e^2} &= \frac{r_j(1+e\cdot\cos(\theta_j-\omega))}{1-e^2} \Leftrightarrow \\ r_i + r_i e\cdot\cos(\theta_i-\omega) &= r_j + r_j e\cdot\cos(\theta_j-\omega) \Leftrightarrow \\ r_i - r_j &= e(r_j\cdot\cos(\theta_j-\omega) - r_i\cdot\cos(\theta_i-\omega))\end{aligned}$$

Some simplifications can be applied:
$C_{ij}\cdot\cos(\alpha_{ij}-\omega) = r_j\cdot\cos(\theta_j-\omega) - r_i\cdot\cos(\theta_i-\omega)$ and
$\alpha_{ij} = \operatorname{atan2}(B_{ij}, A_{ij})$ with
$C_{ij} = \sqrt{A_{ij}^2 + B_{ij}^2}$, $A_{ij} = \cos(\theta_j)r_j - \cos(\theta_i)r_i$ and $B_{ij} = \sin(\theta_j)r_j - \sin(\theta_i)r_i$.
Hence:

$$r_i - r_j = e(C_{ij}\cdot\cos(\alpha_{ij}-\omega)) \Leftrightarrow \frac{r_i-r_j}{C_{ij}\cdot\cos(\alpha_{ij}-\omega)} = e$$

$$\boxed{e = \frac{r_i-r_j}{C_{ij}\cdot\cos(\alpha_{ij}-\omega)}} \tag{21}$$

Next, we obtain ω by equating expression 21 with itself with two different index pairs ij and ik with $k \in \{1,2,3\}$ and $i \neq j \neq k \neq i$ as follows:

$$\begin{aligned}\frac{r_i-r_j}{C_{ij}\cdot\cos(\alpha_{ij}-\omega)} &= \frac{r_i-r_k}{C_{ik}\cdot\cos(\alpha_{ik}-\omega)} \Leftrightarrow \\ (r_i-r_j)\cdot(C_{ik}\cdot\cos(\alpha_{ik}-\omega)) &= (r_i-r_k)\cdot(C_{ij}\cdot\cos(\alpha_{ij}-\omega)) \Leftrightarrow \\ \frac{C_{ik}(r_i-r_j)}{C_{ij}(r_i-r_k)} &= \frac{\cos(\alpha_{ij}-\omega)}{\cos(\alpha_{ik}-\omega)}\end{aligned}$$

Some simplifications can be applied:
$D + E\cdot\tan(\alpha_{ik}-\omega) = \frac{\cos(\alpha_{ij}-\omega)}{\cos(\alpha_{ik}-\omega)}$ with
$D = \cos(\alpha_{ij}-\alpha_{ik})$, $E = -\sin(\alpha_{ij}-\alpha_{ik})$ and also $F = \frac{C_{ik}(r_i-r_j)}{C_{ij}(r_i-r_k)}$.
Hence:

$$F = D + E\cdot\tan(\alpha_{ik}-\omega) \Leftrightarrow -\arctan\left(\frac{F-D}{E}\right) + \alpha_{ik} = \omega$$

With the final simplification of:
$G = \frac{F-D}{E}$
We end up with:

$$\boxed{\omega = \alpha_{ik} - \arctan(G)} \tag{22}$$

As a result, the characteristic parameters a, e and ω of elliptical and hyperbolic trajectories can simply be determined starting with ω from equation 22 then putting that result into equation 21 to determine e and finally putting both values into equation 20 to determine a.
This mathematical model can be used to characterize ellipses and hyperbolas, however there are two special cases where it does not work, namely circular orbits and parabolic trajectories. In the special case of a circular orbit with $e = 0$, all three measured radii

posses the same value $r_1 = r_2 = r_3$. This causes the definition of $F = \frac{C_{ik}(r_i - r_j)}{C_{ij}(r_i - r_k)}$ to be undefined due to zero division, which causes ω to be undefined. Incidentally, no argument of periapsis ω can be defined for a circle as there is no closest point to the center. A circular orbit can simply be expressed using its semi major axis as $r(\theta) = a$. With one known radius value r_i the semi major axis is already sufficiently defined:

$$\boxed{a = r_i} \tag{23}$$

For an orbit to be circular it is necessary that all radii measurements r_1, r_2 and r_3 are exactly the same value wise.
In the special case of parabolic trajectories with $e = 1$, multiple expressions (e.g. 19 and 20) with e in the denominator do not hold up due zero division. Mentioned previously in section 2.1, parabolas can be modeled with equation 4 which is characterized by p and ω. For determining these parameters we only need two distinct measurements where two radii r_i for $i \in \{1, 2\}$ and two angles θ_i for $i \in \{1, 2\}$ are known. The derivation starts by putting values of r_i and θ_i into equation 4, which can be rearranged to obtain p subject to ω:

$$\begin{aligned} r_i &= \frac{p}{1 + \cos(\theta_i - \omega)} \Leftrightarrow \\ r_i \cdot (1 + \cos(\theta_i - \omega)) &= p \\ \boxed{p = r_i \cdot (1 + \cos(\theta_i - \omega))} & \end{aligned} \tag{24}$$

Next, we determine ω by inserting expression 24 with index i into expression 4 with r_j and θ_j put into place for $j \in \{1, 2\}$ and $i \neq j$ as follows:

$$\begin{aligned} r_j &= \frac{r_i(1 + \cos(\theta_i - \omega))}{1 + \cos(\theta_j - \omega)} \Leftrightarrow \\ r_j + r_j \cdot \cos(\theta_j - \omega) &= r_i + r_i \cdot \cos(\theta_i - \omega) \Leftrightarrow \\ r_j - r_i &= r_i \cdot \cos(\omega - \theta_i) - r_j \cdot \cos(\omega - \theta_j) \end{aligned}$$

Some simplification can be applied:
$\sqrt{I^2 + H^2} \cdot \cos(\omega + \beta) = r_i \cdot \cos(\omega - \theta_i) - r_j \cdot \cos(\omega - \theta_j)$ with
$I = r_i \cdot \cos(\theta_i) - r_j \cdot \cos(\theta_j)$, $H = r_i \cdot \sin(\theta_i) - r_j \cdot \sin(\theta_j)$ and
$\beta = -\operatorname{atan2}(H, I)$.
Hence:

$$\begin{aligned} r_j - r_i &= \sqrt{I^2 + H^2} \cdot \cos(\omega + \beta) \Leftrightarrow \\ \arccos\left(\frac{r_j - r_i}{\sqrt{I^2 + H^2}}\right) - \beta &= \omega \end{aligned}$$

With the final simplification of:
$J = \frac{r_j - r_i}{\sqrt{I^2 + H^2}}$
We end up with:

$$\boxed{\omega = \arccos(J) - \beta} \tag{25}$$

As a result, the characteristic parameters p and ω of parabolic trajectories can simply be determined starting with ω from equation 25 then putting that result into equation 24 to determine p.
Now, that mathematical models for determining elliptical, hyperbolic, circular and parabolic trajectories have been established, we can define a generalized process which determines a parametric trajectory from three measurements without the prior knowledge which type of trajectory and thus model it is going to be.
Before doing so it should be noted that some expressions in these models contain singularities which could cause problems due to zero division. In the ellipse/hyperbola model these are contained within equation 21 and expressions for F and G. In the parabolic model it occurs within the expression for J. These singularities only appear if certain

values are exactly identical. Due to the application of these models in a physical, rather than purely mathematical setting and because of their input values which possess a large mantissa and are obtained by camera systems with limited accuracies, singularities are very unlikely to occur. For this reason we simply chose to ignore them as they will have no stake in the average use case. If a singularity were to appear during calculation, it may be averted by permutating indices of measurements and performing a recalculation.
Once again we start off with three radii r_1, r_2 and r_3 and their corresponding angles θ_1, θ_2 and θ_3 as a basis for the following steps:

1. Check if $r_1 = r_2 = r_3$. If yes, assume a circular trajectory and proceed with step 5.
2. Assume an elliptical/hyperbolic trajectory. Calculate all intermediate parameters beforehand using radii and angle values. Determine ω using equation 22.
3. Determine e subject to ω. If $|e| = 1$, assume a parabolic trajectory and proceed with step 6.
4. Determine a subject to e and ω using equation 20. An **elliptical/hyperbolic** trajectory is now sufficiently defined.
   Done.
5. Determine a using equation 23. A **circular** trajectory is now sufficiently defined.
   Done.
6. Determine ω using equation 25.
7. Determine p subject to ω using equation 24. A **parabolic** trajectory is now sufficiently defined.
   Done.

4.2 Transferring the parametric model into three dimensions

The previously established model describes trajectories in a two dimensional plane. Positions are simply referred to as radius or magnitude r_i (a real number) and θ_i (the angle between the radius line and the X-axis). The goal of this section is to expand this parametric model into a three dimensional environment to be applicable in reality. For this purpose, we define and elaborate a three dimensional geometrical context.

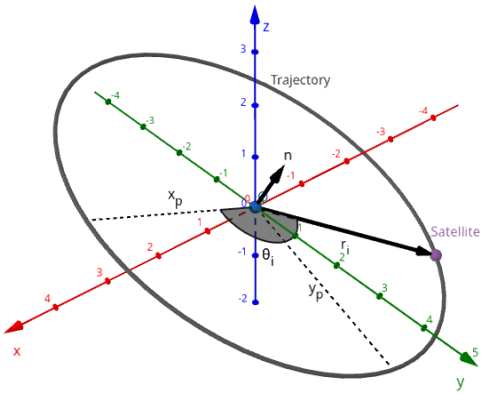


Figure 12: The parametric trajectory in the three dimensional ECI coordinate system.

Illustrated above is a trajectory in three dimensions. It still resides completely within a two dimensional plane which is characterized solely by an orthonormal vector n as the origin of the coordinate system always lies within the plane. Positions in this context are also referred to as r_i with the difference being that they are vectors. The definition of angles θ_i is slightly more complex. Both x_p and y_p are defined as projected unit vectors

$x_0 = (1, 0, 0)$ and $y_0 = (0, 1, 0)$ in the orbital plane. For visual purposes their lengths are not properly represented.
The parametric model relies on θ_i lying in the range of $0 \leq \theta_i \leq 2\pi$ or $-\pi \leq \theta_i \leq \pi$ which can not be calculated using $\arccos(x)$ of the scalar product of two vectors. A more suitable trigonometric function is $\text{atan2}(y, x)$ as it can determine an angle in the range of $-\pi \leq \theta \leq \pi$ in the correct quadrant as the sign of both inputs is known [15]. With these aspects a formula for θ_i can be derived. We start off with the definitions of x_p and y_p:

$$\begin{aligned} x_p &= n \times x_0 \\ y_p &= n \times y_0 \end{aligned} \tag{26}$$

These projected vectors can be seen as X- and Y-axis for the two dimensional parametric model. Next, apply scalar products to r_i to determine its fractions x and y relative to x_p and y_p:

$$\begin{aligned} x &= r_i \cdot x_p \\ y &= r_i \cdot y_p \end{aligned} \tag{27}$$

With $\theta_i = \text{atan2}(y, x)$ we construct the following formula

$$\boxed{\theta_i = \text{atan2}(r_i \cdot (n \times x_0), r_i \cdot (n \times y_0))} \tag{28}$$

which satisfies the requirement of $-\pi \leq \theta_i \leq \pi$. Henceforth, this adaption makes the parametric model usable in a three dimensional environment, as angles and magnitudes can now be determined from vectors for the two dimensional model.
Lastly, a method for determining the orthonormal plane vector n has to be established. A measurement series of satellite positions $r_1, r_2, r_3, ..., r_n$ may contain numerous entries. It must contain at least two entries so that the cross product between two vectors can be formed as follows:

$$\boxed{n = \frac{r_i \times r_j}{|r_i| \cdot |r_j|}} \tag{29}$$

In principle, all vectors of a series should lie exactly within the plane, such that it should not matter which two vectors are chosen to form the cross product. Due to measurement errors caused by various factors this may not be the case. Only the two vectors spanning the plane will fit perfectly, whereas the others might deviate slightly.

4.3 Orbit determination with two cameras

The previous two sections have established models under the assumption that r_i vectors are readily available for computation, but as of yet we have no methods for determining them. The only vectors which are readily available are position vectors p_i and view vectors v_i. One approach for determining satellite positions is to use a two camera setup as illustrated below:

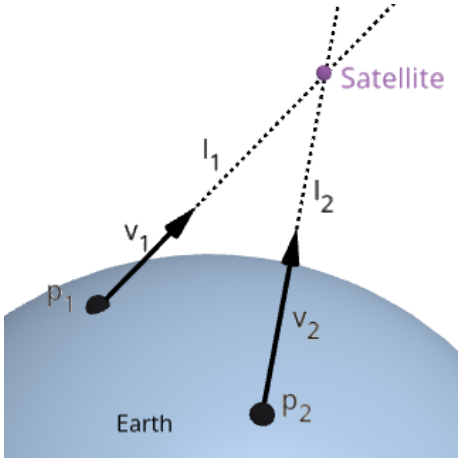


Figure 13: The parametric trajectory in the three dimensional ECI coordinate system.

Hereby, two cameras in different locations take an image of the satellite simultaneously. Immediately known are their position and view vectors. These properties form two lines l_1 and l_2 defined as

$$\begin{aligned} l_1(t) &= p_1 + t \cdot v_1 \quad t \in \mathbb{R} \\ l_2(s) &= p_2 + s \cdot v_2 \quad s \in \mathbb{R} \end{aligned} \tag{30}$$

which should intersect each other in a mathematical setting. However, in a physical setting where cameras are limited in accuracy, an exact intersection is unlikely to occur. A feasible guess for satellite location would be the closest approach between the two lines, more specifically the point in the middle of l_1 and l_2 where they approach each other the closest.
Fitzpatrick presents a readily available solution. At their points of closest approach $l_2(s) - l_1(t)$ is minimal in magnitude and perpendicular both to v_1 and v_2 meaning that scalar products between view vectors and $l_2(s) - l_1(t)$ should be zero:

$$\begin{aligned} 0 &= v_1 \cdot (l_2(s) - l_1(t)) \\ 0 &= v_2 \cdot (l_2(s) - l_1(t)) \end{aligned} \tag{31}$$

This expression can be rearranged into a system of linear equations yielding solutions for t and s hereby defined as t_{min} and s_{min} where the distance between the lines is at their minimum.[16]
As there is no intersection the satellite position r_i can not be determined exactly. However, as feasible value we define r_i as the point located in the middle between l_1 and l_2 at their minimum distance as follows:

$$\boxed{r_i = l_1(t_{min}) + \frac{1}{2}(l_2(s_{min}) - l_1(t_{min}))} \tag{32}$$

Thereby, we obtain a method for full orbit determination using two cameras which can be formulated as a process:

1. Measure position vectors p_i and v_i of two observing cameras for three points in time resulting in six measured vectors total.

2. Determine three radii vectors r_1, r_2 and r_3 using equation 32. The system of linear equations resulting from expression 31 needs to be solved beforehand to obtain t_{min} and s_{min} for each observation.

3. Chose two radii vectors and determine the plane normal vector n using equation 29.

4. With n established, determine the angle θ_i for each radii vector r_i using equation 28.

5. The values of r_i will now be referred to as magnitudes of their corresponding radii vectors. Magnitudes r_i and angles θ_i serve as input for the process defined in section 4.1 which yields a parametric model.
   Done.

4.4 Gaussian method of orbit determination

The Gaussian method of orbit determination is an algorithm originally developed by Gauss that requires only a single camera, thus a less complex setup than the previous approach. Curtis provides the complete algorithm, as in the initial idea, the derivation and the final process which can be used for computation. In the following, we are only going have a look at the idea and the process as the derivation of underlying equations is of no particular relevance for the scope. Furthermore, we are going to apply some modifications to the process, as we can use some previously established solutions to reduce the number of steps. We start off with geometric context and the initial idea, illustrated in the figure below:

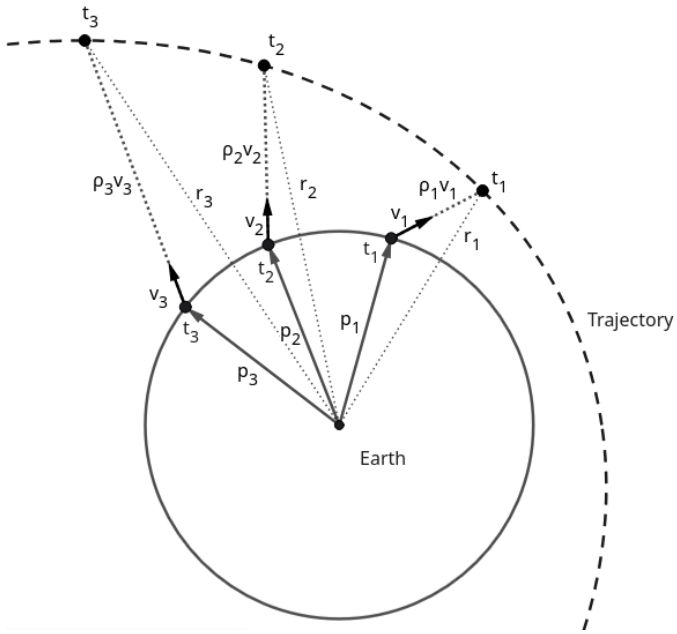


Figure 14: Initially known and unknown properties of Gaussian orbit determination depicted in a two dimensional representation.

Initially, three observations at three different time stamps t_1, t_2 and t_3 from a fixed location on Earth are taken, which yield three position vectors p_1, p_2 and p_3 as well as three view vectors v_1, v_2 and v_3. Unknown vectors are the satellite positions or radii vectors r_1, r_2 and r_3 similar to section 3.1. They can be described using the following relations

$$r_1 = p_1 + \rho_1 v_1 \qquad r_2 = p_2 + \rho_2 v_2 \qquad r_3 = p_3 + \rho_3 v_3 \tag{33}$$

with $\rho_i \in \mathbb{R}$ for $i \in \{1, 2, 3\}$. Through this, we reduce the unknown parameters to three real numbers. The goal of this algorithm is to determine fitting solutions for the unknowns so that the observations match a physical orbit around planet Earth considering gravitational parameters. This solution can be calculated using the following procedure as elaborated by Curtis:

1. Determine time intervals τ_1, τ_3 and τ from t_1, t_2 and t_3 as follows:

$$\tau_1 = t_1 - t_2 \qquad \tau_3 = t_3 - t_2 \qquad \tau = \tau_3 - \tau_1 \tag{34}$$

2. Determine the cross products m_1, m_2 and m_3 from view vectors v_1, v_2 and v_3 as follows:

$$m_1 = v_2 \times v_3 \qquad m_2 = v_1 \times v_3 \qquad m_3 = v_1 \times v_2 \tag{35}$$

3. Determine the scalar product D_0 as follows:

$$D_0 = v_1 \cdot m_1 \tag{36}$$

4. Calculate the nine scalar products D_{ij} for $i, j \in \{1, 2, 3\}$ as follows:

$$\begin{aligned} D_{11} &= p_1 \cdot m_1 & D_{12} &= p_1 \cdot m_2 & D_{13} &= p_1 \cdot m_3 \\ D_{21} &= p_2 \cdot m_1 & D_{22} &= p_2 \cdot m_2 & D_{23} &= p_2 \cdot m_3 \\ D_{31} &= p_3 \cdot m_1 & D_{32} &= p_3 \cdot m_2 & D_{33} &= p_3 \cdot m_3 \end{aligned} \tag{37}$$

5. Calculate real number parameters A and B using the following expressions:

$$\begin{aligned} A &= \frac{1}{D_0}\left(-D_{12}\frac{\tau_3}{\tau} + D_{22} + D_{32}\frac{\tau_1}{\tau}\right) \\ B &= \frac{1}{6D_0}\left(D_{12}\left(\tau_3^2 - \tau^2\right)\frac{\tau_3}{\tau} + D_{32}\left(\tau^2 - \tau_1^2\right)\frac{\tau_1}{\tau}\right) \end{aligned} \tag{38}$$

6. Determine the scalar products E and p_2^2 as follows:

$$E = p_2 \cdot v_2 \qquad p_2^2 = p_2 \cdot p_2 \tag{39}$$

7. Determine the real number parameters a, b and c using the following expressions:

$$a = -\left(A^2 + 2AE + p_2^2\right) \qquad b = -2\mu B(A + E) \qquad c = -\mu^2 B^2 \tag{40}$$

   Hereby, μ represents the gravitational parameter, the product of Earth's mass and the universal gravity constant.

8. An eighth degree polynomial equation containing the parameters a, b and c is defined as follows:

$$x^8 + ax^6 + bx^3 + c = 0 \tag{41}$$

   Of this polynomial, a real number root w larger than zero as solution for x must be determined. If there is more than one solution w it means that there is more than one physically feasible trajectory. The entire algorithm must be started over from step 1 exchanging one of the measurements for an additional one until a single solution w has been determined. For computation, Curtis proposes using Newton's method and plotting the polynomial as function to determine a proper initial estimate of x. We propose using a solver like NumPy-roots which determines the roots of any polynomial [18] and works out of the box.

9. Determine the unknown parameters ρ_1, ρ_2 and ρ_3 using w with the following expressions:

$$\begin{aligned} \rho_1 &= \frac{1}{D_0}\left(\frac{6\left(D_{31}\frac{\tau_1}{\tau_3} + D_{21}\frac{\tau}{\tau_3}\right)w^3 + \mu D_{31}\left(\tau^2 - \tau_1^2\right)\frac{\tau_1}{\tau_3}}{6w^3 + \mu\left(\tau^2 - \tau_3^2\right)} - D_{11}\right) \\ \rho_3 &= \frac{1}{D_0}\left(\frac{6\left(D_{13}\frac{\tau_3}{\tau_1} - D_{23}\frac{\tau}{\tau_1}\right)w^3 + \mu D_{13}\left(\tau^2 - \tau_3^2\right)\frac{\tau_3}{\tau_1}}{6w^3 + \mu\left(\tau^2 - \tau_1^2\right)} - D_{33}\right) \\ \rho_2 &= A + \frac{\mu B}{w^3} \end{aligned} \tag{42}$$

10. Use the previously defined equations 33 to determine r_1, r_2 and r_3 from ρ_1, ρ_2 and ρ_3 and already known vectors.

[17]
From here on onward the Gauss algorithm as presented by Curtis contains three additional steps to establish orbital properties from r_1, r_2 and r_3 determined in step 10. However, we have already established a method to determine orbital properties which we reuse to formulate a single final step as follows:

11. Use the orbit determination procedure established in section 4.3 starting from step 3 as r_1, r_2 and r_3 are already known. This yields vector n and a parametric model. **Done.**

4.5 Simplified Gaussian method of orbit determination

The previously presented Gaussian method determines viable parameters to form all types of parametric trajectories that fit observations to physical orbits. Due to orbits of satellites at lower altitudes above the Earth being mostly circular, we propose a method that always assumes a circular model. This way we only need to determine the plane normal vector n and the semi major axis a. The aspect that we want to keep is the use of a single camera so that the setup is very much similar to figure 14, except that one measurement less is needed. Known values are the position vectors p_1, p_2, the view vectors v_1, v_2, whereupon they are related similar to expression 33 so that r_1, r_2, ρ_1 and ρ_2 are unknowns. Furthermore, the time stamps t_1, t_2 of measurement recordings are supposed to be known.
We start off with the circularity condition, meaning that all radii vectors r_i must possess the same magnitude which formulates as follows:

$$\boxed{|r_1| = |r_2| \Leftrightarrow \sqrt{(p_1 + \rho_1 v_1)^2} = \sqrt{(p_2 + \rho_2 v_2)^2}} \tag{43}$$

Next, the orbital period T subject to the semi major axis a and the gravitational parameter μ can be expressed as follows:

$$T = \frac{2\pi}{\sqrt{\mu}} a^{\frac{3}{2}} = \frac{2\pi}{\sqrt{\mu}} |r_i|^{\frac{3}{2}} \tag{44}$$

[3]
Knowing that $a = |r_1| = |r_2|$, the value of a can be substituted for the magnitude of any radius vector r_i. Next, the angle φ with $0 \leq \varphi \leq \pi$ between r_1 and r_2 can be described using the arccos as follows:

$$\varphi = \arccos\left(\frac{r_1 \cdot r_2}{|r_1| \cdot |r_2|}\right) \tag{45}$$

This angle describes the arc of the circular orbit which the satellite traverses between time stamps t_1 and t_2 under the condition that $t_2 - t_1 \leq \frac{T}{2}$. This restriction needs to be in place, because the output range of arccos does not cover a full circle. This relation between time stamps and arc formulates as:

$$t_2 - t_1 = \Delta t = \frac{\varphi}{2\pi} T \tag{46}$$

Equations 44 and 45 can be inserted into 46 to form the following expression

$$\frac{\arccos\left(\frac{r_1 \cdot r_2}{|r_1| \cdot |r_2|}\right)}{2\pi} \cdot \frac{2\pi}{\sqrt{\mu}} |r_1|^{\frac{3}{2}} = \Delta t \Leftrightarrow$$
$$\arccos\left(\frac{r_1 \cdot r_2}{|r_1|^2}\right) \cdot |r_1|^{\frac{3}{2}} = \Delta t \sqrt{\mu}$$

which yields the time condition subject to ρ_1 and ρ_2 formulated as:

$$\boxed{\arccos\left(\frac{(p_1+\rho_1 v_1)\cdot(p_2+\rho_2 v_2)}{|(p_1+\rho_1 v_1)|^2}\right)\cdot|(p_1+\rho_1 v_1)|^{\frac{3}{2}} = (t_2-t_1)\sqrt{\mu}} \tag{47}$$

With circularity and time conditions established, we can formulate a full orbit determination process as follows:

1. Measure position vectors p_1, p_2 and view vectors v_1, v_2 of two time stamps t_1 and t_2. A crucial requirement for this algorithm to work is that $t_2 - t_1 \leq \frac{T}{2}$ meaning that observation arcs can not take longer than half the orbital period.

2. A real number solution for ρ_1 and ρ_2 must be computed which satisfies the time condition 47 constrained by the circularity condition 43. This task can not be accomplished analytically. For this reason we propose the use of a sophisticated solver like scipy.optimize.root which allows for modeling more complex functions to be solved numerically [19]. This yields the two values ρ_1 and ρ_2.

3. Calculate r_1 and r_2 using expressions 33. Obtain the plane normal vector n using equation 29. The semi major axis a equates radii vector magnitudes $a = |r_1| = |r_2|$. **Done.**

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